



TEAM 6
AWD



BIG DATA

MEDICAL APPOINTMENT SYSTEM

WHAT IS BIG DATA?



Big data refers to extremely large and diverse collections of structured, unstructured, and semi-structured data that continues to grow exponentially over time. These datasets are so huge and complex in **volume**, **velocity**, and **variety** that traditional data management systems cannot store, process, and analyze them.

BIG DATA IN HEALTHCARE



Healthcare data sources include genomics data sets, electronic health records, patient data from wearables and biosensors, and third party data from insurance claims.

Many medical devices are part of the internet of things and these **IoT devices collect** and transmit **large amounts of data about patients** from a variety of sources.

Examples include patient vitals from monitors, appointment histories, prescription records, lab results, medical imaging, fitness tracker data, even patient feedback from surveys or social media.

There is often an urgency involved in the use of medical data and that's where processing velocity comes into play, like data that reveals drug interactions



BIG DATA PROPOSAL

IN THE MEDICAL APPOINTMENT WEB APPLICATION

● SMART SCHEDULING WITH PREDICTIVE ANALYTICS

Analyze historical appointment data to **predict no shows based on patient patterns**. The system learns which patients are likely to miss appointments and automatically **sends them extra reminders** or books them at better times. It can also identify peak booking periods and optimize doctor schedules accordingly.

Use **machine learning to forecast appointment demand** by day, week, season so the clinic can adjust staffing levels and availability dynamically. This reduces wait times and improves resource utilization.

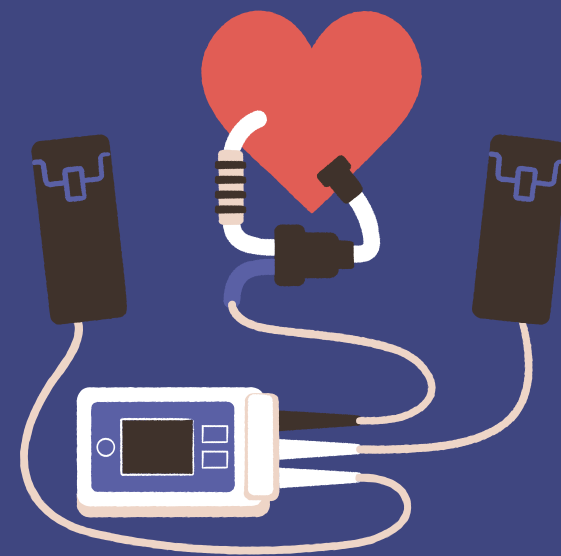




INTEGRATION WITH WEARABLE HEALTH DEVICES



Allow patients to **connect their fitness trackers or health monitors to their appointment profile**. The system analyzes their health data trends and can automatically suggest booking checkups when concerning patterns emerge.



For example if a patient's connected blood pressure monitor shows consistently high readings, the system sends them a notification to schedule an appointment and even pre selects cardiology specialists.





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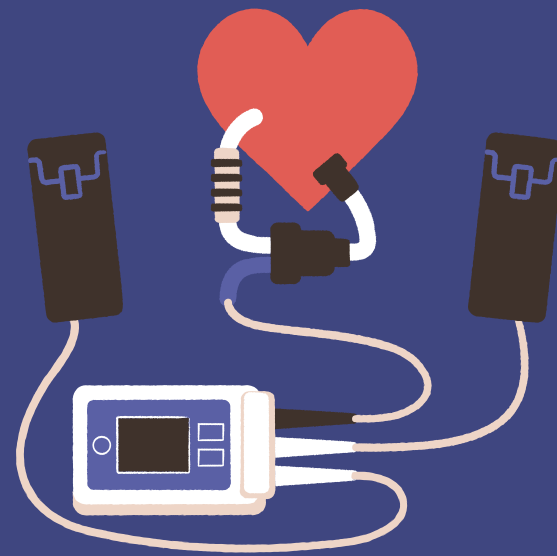
DATA MINING

MEDICAL APPOINTMENT SYSTEM



WHAT IS DATA MINING

Data mining is the process of discovering useful patterns, trends, and relationships within large datasets using statistical techniques, algorithms, and machine learning. Unlike Big Data, which focuses on managing large volumes of information, data mining focuses on extracting valuable knowledge from stored data.



DATA MINING

EXAMPLES

Netflix uses data mining to analyze what users watch, how long they watch, when they pause, and which titles they complete. It trains recommendation models to suggest shows like “Because you watched...”, using collaborative filtering and viewing behavior patterns.



Amazon performs market basket analysis to discover which products are frequently bought together in the same order. Using association rules, it powers features such as “Frequently bought together” and “Customers who bought this item also bought...”, boosting



Platforms like YouTube mine play history, skips, likes, and searches to learn user preferences. They apply machine learning and collaborative filtering to group similar users and recommend songs or videos that increase engagement and watch time.



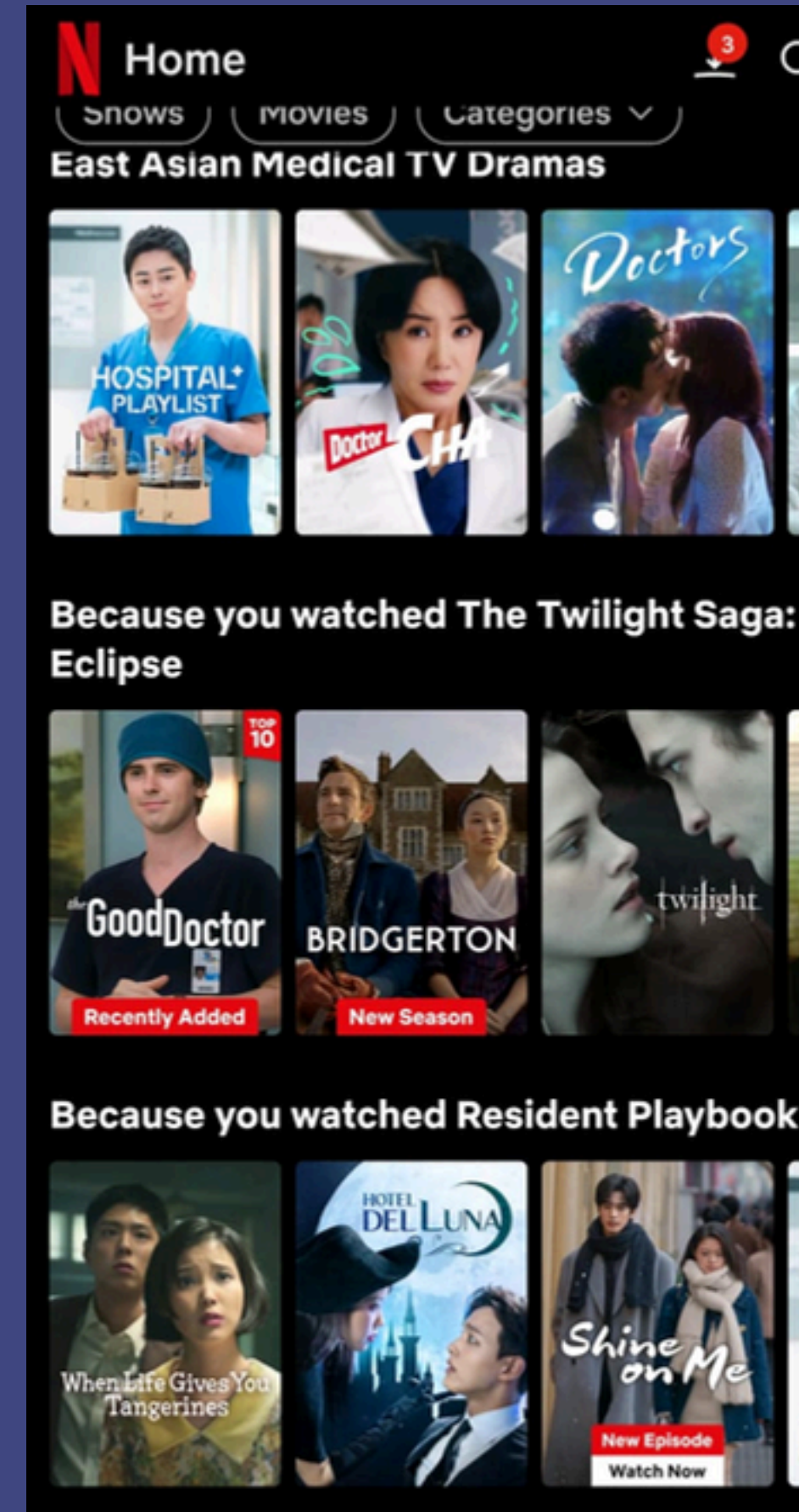
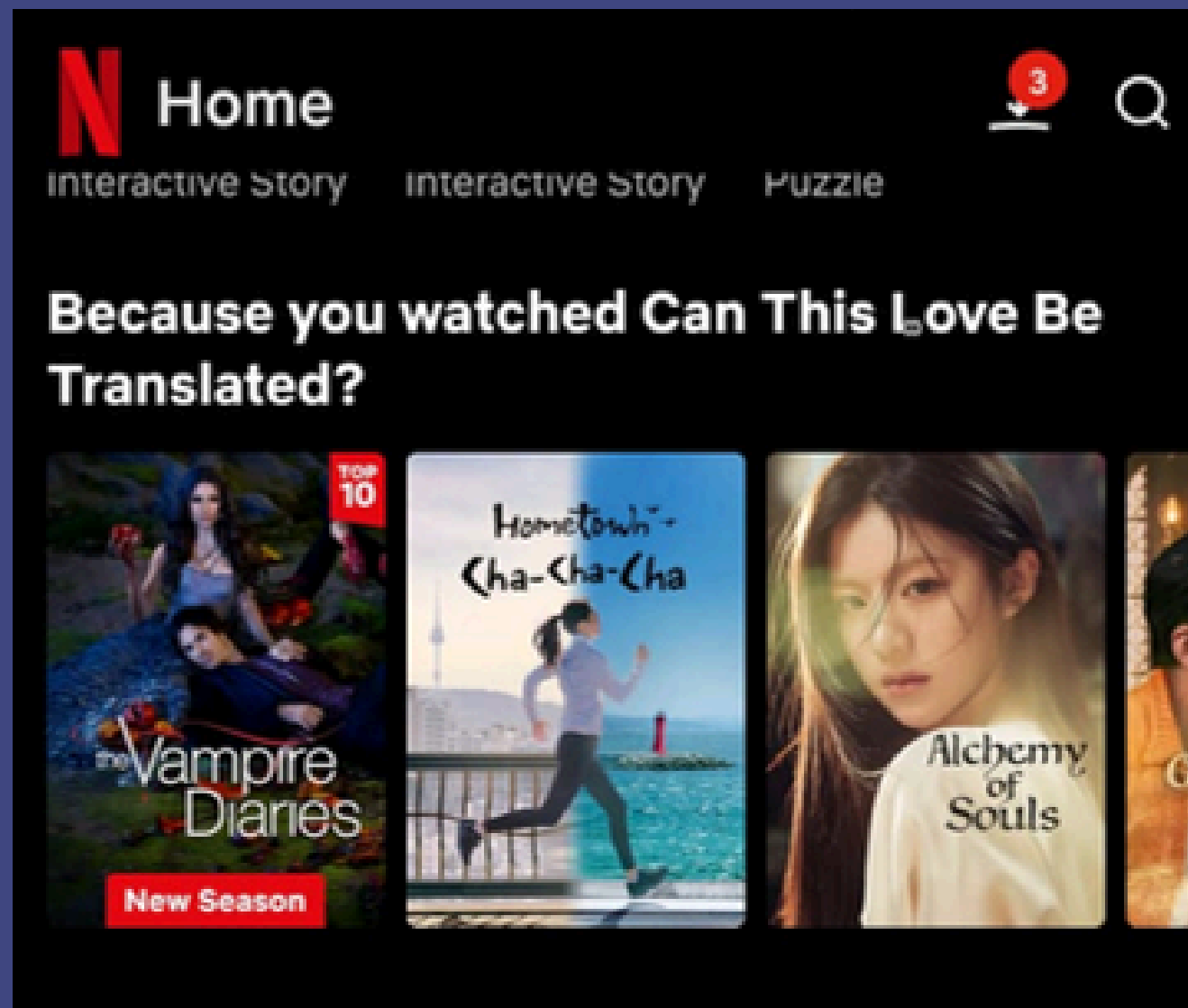
Walmart uses data mining on large sales and weather datasets to understand which products sell together, at which times, and under which conditions like storms or heat waves. These insights help optimize inventory, store layouts, and targeted promotions across its stores.



DATA MINING

EXAMPLES

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DATA MINING

● MEDICAL APPOINTMENT SCHEDULING WEBSITE

In a web-based medical appointment scheduling system, data sources include **appointment histories**, **patient demographics**, attendance and no-show records, medical specialties consulted, most frequently used appointment times, cancellations, rescheduling, and average appointment duration.

Using data mining techniques, the system can **analyze this historical data to identify patient behavior patterns**. For example, it can detect profiles of patients who frequently miss appointments, those who cancel often, or those who prefer certain days and times.



DATA MINING

IN THE MEDICAL APPOINTMENT WEB APPLICATION

One application example is predicting no-shows. Using association rules and classification models, the system can determine which factors most influence a patient's absence from their appointment (time, day, specialty, prior history). With this information, the system can take preventative measures such as sending personalized reminders and requesting confirmations.

Data mining also allows for patient segmentation based on their system usage patterns. For example, it can identify frequent patients, occasional patients, or patients with chronic illnesses requiring regular checkups. This segmentation helps improve appointment scheduling and provide more personalized care.





**iTHANK
YOU!**